

SHELAGONTSY, Russian Federation

WMO: 243290

Lat: 66.250N

Lon: 114.283E

Elev: 774

StdP: 14.29

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-65.3	-61.4	-67.2	0.1	-65.1	-63.4	0.1	-61.0	16.0	-3.0	12.4	-9.7	0.7	230	0.772

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.7	83.1	63.3	78.5	61.3	74.1	59.0	65.3	77.9	63.1	75.1	60.9	71.7	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	82.0	69.6	58.3	74.9	67.7	55.8	68.3	65.2	30.7	77.8	29.0	75.4	27.4	71.9	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	11.8	10.1	DB	-65.6	88.8	3.7	3.3	-68.3	91.2	-70.5	93.1	-72.5	95.0	-75.2	97.4	(4)
(5)			WB	-65.4	70.1	3.7	2.3	-68.1	71.8	-70.2	73.2	-72.3	74.5	-74.9	76.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-37.7	-30.9	-9.7	14.7	34.9	53.6	58.6	51.1	36.2	12.9	-19.5	-36.6
	DBStd	37.18	16.50	15.16	15.74	12.53	10.13	8.44	7.17	6.72	8.07	14.67	15.89	14.64
	HDD50	14842	2720	2266	1851	1059	476	56	11	67	417	1150	2085	2684
	HDD65	19798	3185	2686	2316	1509	935	352	220	432	864	1615	2535	3149
	CDD50	553	0	0	0	0	7	163	279	101	3	0	0	0
	CDD65	33	0	0	0	0	0	9	22	2	0	0	0	0
	CDH74	922	0	0	0	0	3	254	579	84	2	0	0	0
	CDH80	251	0	0	0	0	0	62	171	18	0	0	0	0
Wind	WSAvg	2.7	1.7	1.8	2.6	3.7	4.2	3.6	3.2	3.0	3.0	2.8	2.0	1.4
Precipitation	PrecAvg	11.5	0.4	0.3	0.4	0.5	1.1	1.8	2.2	1.8	1.1	0.9	0.6	0.4
	PrecMax	17.4	0.9	0.6	0.9	1.7	2.0	3.9	6.8	4.6	2.2	1.5	1.3	1.0
	PrecMin	6.9	0.1	0.0	0.1	0.0	0.1	0.2	0.0	0.3	0.2	0.4	0.2	0.0
	PrecStd	2.6	0.2	0.2	0.2	0.3	0.5	0.9	1.5	1.1	0.6	0.3	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	17.9	37.2	52.2	72.9	87.1	89.3	82.2	65.8	43.1	28.4	9.7
		MCWB	15.7	16.6	31.3	40.0	52.4	63.4	66.8	64.0	52.7	38.7	26.9	8.5
	2%	DB	6.9	9.1	29.8	45.7	64.5	81.4	84.7	75.1	57.5	38.7	21.2	2.1
		MCWB	6.1	8.0	25.4	36.3	48.9	61.8	64.2	60.3	49.0	35.9	19.8	1.2
	5%	DB	-2.3	0.7	23.8	40.3	58.1	76.8	80.6	70.8	53.2	35.1	12.1	-7.0
		MCWB	-2.8	-0.1	20.7	33.4	45.6	59.3	62.5	57.9	45.6	32.9	11.2	-7.4
	10%	DB	-13.0	-7.0	17.7	36.2	52.5	72.1	76.5	66.4	49.6	31.8	5.4	-13.6
		MCWB	-13.3	-7.5	15.3	31.0	42.3	56.9	61.2	55.9	43.9	29.7	4.7	-13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	16.4	31.5	40.3	53.9	66.6	69.2	65.2	55.0	39.3	27.3	8.8
		MCDB	16.9	18.1	37.0	51.0	70.2	78.9	82.8	77.1	61.3	42.5	28.2	9.7
	2%	WB	5.9	8.1	25.5	37.1	49.7	63.4	66.6	62.1	49.9	36.3	19.7	1.2
		MCDB	6.8	9.2	29.6	44.3	61.9	77.2	79.5	71.9	54.5	38.4	21.2	2.2
	5%	WB	-2.9	0.0	20.8	34.1	46.1	61.0	64.4	59.6	47.1	33.0	11.3	-7.4
		MCDB	-2.4	0.8	23.9	39.7	57.1	74.2	76.5	67.9	52.3	34.9	12.3	-7.1
	10%	WB	-13.3	-7.5	15.3	31.1	42.8	58.3	62.5	57.1	44.4	29.8	4.8	-13.9
		MCDB	-13.1	-6.9	17.6	36.1	51.9	69.9	73.9	65.0	48.7	31.6	5.6	-13.6
Mean Daily Temperature Range	5% DB	MDBR	16.8	22.7	34.1	30.8	20.8	22.5	24.7	23.6	17.8	16.8	17.9	15.8
		MCDBR	25.3	28.1	32.6	31.8	29.0	32.2	34.2	33.9	26.4	16.6	24.3	25.3
		MCWBR	24.5	27.6	28.7	24.0	16.8	15.6	16.1	19.6	17.6	12.8	23.2	24.8
	5% WB	MCDBR	25.0	28.1	31.9	28.5	27.2	28.3	28.9	27.8	23.2	14.4	23.8	25.0
		MCWBR	24.5	27.6	28.2	21.9	16.7	15.0	15.8	17.7	16.3	12.1	23.1	24.5
Clear-Sky Solar Irradiance	taub	0.186	0.237	0.285	0.314	0.344	0.373	0.415	0.419	0.308	0.249	0.178	0.179	
	taud	1.871	2.057	2.075	2.103	2.254	2.387	2.257	2.172	2.482	2.205	1.893	1.879	
	Ebn at Noon	71	196	242	266	270	265	246	228	243	192	70	8	
	Edh at Noon	6	17	29	37	36	33	36	36	20	15	5	1	
All-Sky Solar Radiation	RadAvg	30	266	814	1468	1768	1738	1630	1103	629	271	66	4	
	RadStd	2	14	34	73	97	149	109	133	63	35	5	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page